

Where every number comes from.

Sewershed fuses two independent sources: **Cyvl's AI street-scan** (what the surface looks like) and **Somerville's public GIS** (what's buried under it). Nothing is mocked — this sheet is the exact inventory and which side each layer is read from.

Cyvl street-scan Somerville GIS Derived by Sewershed

Cyvl street-scan AI VISION · VAN-MOUNTED	Somerville GIS PUBLIC · CITY OPEN DATA
Pavement condition 5,080 PCI 0–100 per segment · the dig-readiness driver	Sewer gravity mains 11 layers WATERTYPE, install date, material, diameter, manhole IDs, invert depths
Pavement distresses 84,612 cracking, patching, potholes — typed & located	Combined segments 2,404 carry sewage + stormwater together — what we rank
Above-ground assets 8,254 basins 381 · manholes 841 · ramps 748 · sidewalks 864 · curbs 915 · + trees, hydrants, signals	Catchments 7 sewer drainage areas A · CA · C1 · C2 · S1 · S2 · M
Catch basins 381 location + street-level photo (condition was null)	Storm inlets 3,659
Traffic signs 3,782 MUTCD class + condition	Water mains 8,028
Road markings & striping 7,116	Water-pipe risk 2,061 RiskQuad: Failing / High / Monitor / Low
Street-level imagery per-asset CloudFront photo URL per asset	Pipe age & depth per-pipe from INSTALLDAT + invert elevations

Derived by Sewershed	OUR COMPUTE — JOINS BOTH SOURCES
Dig-readiness score (0–100) 2,404 6 weighted factors: pavement urgency + pipe age + dig depth + asset bundling + network leverage + water co-risk	
Catch-basin condition 381 knowledge distillation on Cyvl imagery	
Cyvl photo → crop to asset marker → Claude Vision labels 112 (teacher) → distilled classifier scores all 381 (student)	

Honest coverage. Cyvl's scan covers the streets the survey van actually drove — about **29 of Somerville's ~106 road-miles**. We load 100% of what Cyvl captured and never invent pavement for unscanned streets; segments without a scan are flagged "not scanned" rather than guessed. The GIS sewer network is city-wide and complete. Pavement layer re-pulled across full city bounds — superseding an earlier export that was clipped to central Somerville.